

DS 310 — Homework 3

Indexing in Practice

Grading: 50 points are graded automatically on Gradescope; the written explanations are graded by hand (4 points per question, 16 points). Assignment total: 66 points.

Overview

In this assignment you run experiments on a database with 500,000 rows and observe how indexes affect query performance. You write code in two places — the four queries in Q3.1 and the three CREATE INDEX statements in Q3.2 — and both are auto-graded. The written explanations in Q3.1, Q3.2, Q3.3, and Q4 are graded by hand (4 points each). Everything else is provided; just run it and record what you observe.

Getting started

Open `YourName_DS310-HW3.ipynb` and place `applications_large.db` in the same folder. Run the **Setup** cell first — it connects to the database and defines `run()`, `time_query()`, and `show_plan()`. Then run the cells in order.

The data

The Application table has 500,000 rows:

```
Application(application_id, study_id, participant_id, submitted_at, status)
```

Q3.1 — Queries without an index

First, time four queries with no extra indexes (other than the automatic primary-key index on `application_id`). Write the four queries below — **do not change the variable names** — then run the timing cell and record the four times.

Code (auto-graded) (20 points)

- `q_point` (5 pts): return the full row for the application with `application_id = 250000`.
- `q_status` (5 pts): return the `COUNT` of applications whose status is `'completed'`.
- `q_ptcp` (5 pts): return all rows for `participant_id = 42`.
- `q_range` (5 pts): return the `COUNT` of applications submitted between `'2024-01-01'` and `'2024-06-01'` (inclusive).

Written (graded by hand — 4 points)

The point lookup on `application_id` is much faster than the others even without an explicit index. Why?

Q3.2 — Queries with indexes

Now add three indexes, then re-time the same queries and observe the speedups.

Code (auto-graded) (10 points)

- Create `idx_status` on `Application(status)`, `idx_participant` on `Application(participant_id)`, and `idx_date` on `Application(submitted_at)`; then commit.

Written (graded by hand — 4 points)

- **A.** Which query saw the largest speedup? Why does indexing help especially for that query?
- **B.** The `status` index gave a smaller speedup than the `participant_id` index. `status` has only 4 distinct values, while `participant_id` has thousands. How does this difference (selectivity) explain the gap?

Q3.3 — The cost of indexes on writes

Indexes speed up reads, but every `INSERT` must also update each index. Run the provided benchmark cell, which times inserting 1,000 rows with and without indexes.

Written (graded by hand — 4 points)

Based on what you observed, describe the trade-off involved in adding indexes to a table. In what situation would you choose not to add an index, even on a frequently queried column?

Q4 — B+ Tree (20 points autograded + 4 points written)

Implement the four missing pieces of a provided B+ Tree (order `m`: internal nodes hold at most `2m` keys; all data lives in linked leaves). Each blank is marked `# YOUR CODE HERE`, and a small test cell under each shows the expected output so you can verify your work before the self-check runs.

Code (auto-graded): `search` (5 pts) returns the value for a key; `range_search` (5 pts) collects (key, value) pairs across the linked leaves; `_split_leaf` (5 pts) splits a full leaf, copying the new leaf's first key up; `_split_internal` (5 pts) splits a full internal node, pushing the middle key up.

Then run the provided full-tree build and the timing experiment (100, 1,000, and 10,000 keys).

Written (graded by hand — 4 points): **A.** How does search time change as `n` grows — is it consistent with $O(\log n)$? **B.** Why does a B+ tree stay shallow even with 10,000 keys?

Q5 — Use of AI (0 points)

Answer the multiple choice question about your use of AI. This question is worth zero points, and your answers do not affect your grades on this or any homework.

Submitting

Run all cells top to bottom (**Kernel** → **Restart & Run All**), make sure the checks pass, then upload `YourName_DS310-HW3.ipynb` to Gradescope. The written answers (Q3.1, Q3.2, Q3.3, Q4) are graded by hand — 4 points each, 16 points total.